Vendor Concentration in Federal Civilian Contracting, FY2008 to FY2025

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GitHub. https://github.com/abh5227/federal-contracting-concentration

Abstract

Studies of how much federal contract money goes to a few firms have focused on the Department of Defense. We found no comparable measure for the civilian agencies, every federal agency other than Defense, such as Veterans Affairs and NASA. We build one from every federal contract action on USAspending from fiscal year (FY) 2008 to FY2025. We split the actions into segments, one agency buying one product in one year, such as Veterans Affairs buying managed health care in FY2025. We score each segment with the Herfindahl-Hirschman Index (HHI), which the Justice Department and the Federal Trade Commission use to measure market concentration. A score above 2,500 counts as highly concentrated. FY2012 to FY2020 is the longest stretch of consistent product codes. Over it, the share of civilian contract dollars in highly concentrated segments rose from an average of 41% in the first three years to 44% in the last three. It has averaged 49% since FY2021. Veterans Affairs health care leads. Three of its segments, worth $44.0 billion together, are each 98% to 100% held by one company, UnitedHealth, McKesson and TriWest. By its competition code, concentrated money was competed about as often as other money. Two findings shape the next phase. Scores swing so much from year to year that many apparent rises look like chance. Part of the swing may come from contracts that commit several years of money at once, which the next phase will test. In addition, 81 large firms appear under two or more IDs in some years, even after each company is rolled up to its parent. That understates concentration in the markets they lead.

1. Introduction

In fiscal year (FY) 2025, Veterans Affairs spent $22.8 billion through contracts on managed health care, and one company received 98% of it. That is one federal market among thousands. Across all of them, the government obligated $793 billion through contracts that year, according to the Government Accountability Office (GAO) [1]. An obligation is a legal commitment to pay. In defense, a few firms now lead most major weapon programs. The number of US aerospace and defense prime contractors, the firms that hold contracts directly with the government, fell from 51 in 1993 to 5 in 2000 [2]. At least one of those five is a prime contractor on 58 of the 78 major weapon systems (74%) in the FY2024 budget of the Department of Defense (DoD) [3]. Concentration matters to the government as a buyer. Few real bidders can mean less price pressure and more dependence on one supplier. A high score is not evidence of wrongdoing, and some purchases have only one capable supplier. This report describes where concentration sits and how it changes. Question 3 will look at what goes with rising concentration, which can point to causes but cannot prove them.

Studies that measure contract concentration directly focus on defense [4, 5, 6]. The closest one compares DoD with all other agencies combined, but only for five product categories over FY2010 to FY2019 [7]. The civilian agencies are every federal agency other than DoD, such as Veterans Affairs, Energy, Homeland Security and NASA. They spent $301 billion through contracts in FY2025, 38% of all contract dollars. We found no published measure that covers them agency by agency and product by product. That is the domain problem. Offices that oversee federal buying cannot tell which civilian markets depend on one or two firms, or whether that dependence is growing.

The analytic problem is to turn tens of millions of transaction rows into one trustworthy score for each agency, product and year, and then to study how those scores move. The project asks three questions.

1. Can USAspending records support a trustworthy concentration score?
2. What common paths does concentration follow over time, such as a steady rise, a steady fall or a one-time jump?
3. What do the agencies and products that came to rely more on a few vendors have in common, such as sole-source awards?

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Sole-source awards are contracts made without competition. This first report answers question 1 and takes a first look at question 2, and question 3 comes in the next phase. By a trustworthy score we mean three things. Our copy of the data matches the published files. Our yearly totals match the government's own totals from GAO. And the results hold steady when we handle problem records, such as reversed entries or returned money, in other reasonable ways. The civilian agencies are the focus. Defense appears beside them as a benchmark, since earlier studies cover it. The measure, its pipeline, our integrity review of the data and our review of documented gaps in the government's data are our own work.

2. Methods

Data. We use USAspending's Award Data Archive [8], which holds contract files for all agencies, in its release of September 6, 2026. The federal fiscal year runs from October 1 to September 30, so FY2025 began on October 1, 2024. The 18 yearly zip files for FY2008 to FY2025 hold 88,753,968 rows and 297 columns, with $10.5 trillion in net obligations. Each row is one contract action, such as a new award, a modification or an order. It records who bought what, from whom and for how much, and whether the award was competed.

Pipeline. The pipeline keeps every row in stages. Our Parquet copy, a compressed column format, is verified against the published zip files. For every file, two independent readers agree on the row count, the exact dollar total and a checksum over every field. Our integrity review [9] then checks every row against 62 rules. It documents many smaller problems and five that would distort the measure. The five are a $13.9 billion entry reversed in the next fiscal year, two $10 billion US Mint rows, negative vendor totals, one placeholder name for many foreign vendors and a placeholder funding agency in the early years. Our yearly totals match GAO's within 0.2% for FY2022 to FY2025, the years GAO publishes exact figures [1, 10, 11, 12].

Segments and the score. A segment is one awarding agency, one four-character product or service code, such as 1510 for fixed-wing aircraft, and one fiscal year. It stands in for one buyer's market for one kind of purchase. A vendor is the parent company ID on the row, or the company's own ID where no parent is given. A joint venture with its own ID, such as BCCG A Joint Venture in Table 2, counts as one vendor, apart from its partner firms. Parent IDs can lag a merger, and the parent field is blank on some rows, so one firm can appear under several IDs (Section 3.1). In the next phase, before the trend analysis, we will check the 100 largest vendors by hand. We will start from the firms whose IDs share a name and merge IDs that belong to one firm. A deobligation, money the government takes back, counts as negative dollars, and each vendor's actions are summed to a net within the segment. The Herfindahl-Hirschman Index (HHI) is a commonly accepted measure of market concentration, and the Justice Department and the Federal Trade Commission use it in merger review [13]. It squares each vendor's percentage share of the segment's dollars and adds them up over P, the set of vendors whose net in the segment is positive:

$$\mathrm{HHI} = \sum_{i \in P} s_i^2,$$

$$s_i = 100 \times \frac{\mathrm{net}_i}{\sum_{j \in P} \mathrm{net}_j},$$

Table 1. HHI rises as the money gathers in fewer hands. The calculation for four example segments of $100 million each, where each share is a vendor's percentage of the segment's dollars.

Vendor shares	Calculation	HHI
Five vendors at 20% each	$5 \times 20^2 = 5 \times 400$	2,000
Four vendors at 25% each	$4 \times 25^2 = 4 \times 625$	2,500
One at 80%, four at 5% each	$80^2 + 4 \times 5^2 = 6{,}400 + 100$	6,500
One vendor at 100%	100^2	10,000
Highly concentrated, 2010 merger guidelines		above 2,500
Highly concentrated, 2023 merger guidelines		above 1,800

The score runs from near 0 for many small vendors to 10,000 for a single vendor, and n equal vendors score 10,000 divided by n (Table 1). The 2010 merger guidelines call a market above 2,500 highly concentrated [14], and the 2023 guidelines lower that line to 1,800 [15].

We use 2,500, the higher line, so every segment we call highly concentrated meets both guidelines. Our main measure is the share of dollars in segments above 2,500. Two segments can score alike for different reasons. A segment with n vendors cannot score below 10,000 divided by n, so one with three or fewer vendors is highly concentrated by arithmetic. One with dozens of vendors scores that high only when one firm dominates the rest. So we report the vendor count beside the score (Table 2), and question 3 will use each line's starting vendor count as a feature. HHI uses shares, so it needs no inflation adjustment. Dollar figures are in current dollars, the amounts as recorded. To compare spending across years, we also give growth after inflation, using the GDP price index, a government measure of how prices change across the whole economy, reported by calendar year [16]. The measure has two limits. It counts prime contractors only, and it looks at one buyer at a time, so a firm can dominate one agency's purchases and still face rivals elsewhere. Both limits can be checked in the next phase (Section 4).

Proposed data handling. This report follows the integrity review's proposed data handling, which the team has not yet made final [9]. It nets the five reversed pairs, holds out the two US Mint rows and leaves the foreign placeholder and rows with no vendor ID out of HHI (Appendices B and C).

Trend lines and the rising rule. A trend line follows one

agency and product code across years. The product code manuals changed in FY2012 [17] and FY2021 [18, 19]. A code that continues may not mean the same thing after a change. The FY2021 manual moved much of IT services out of the old D3 codes into new, narrower ones, and D3 fell from $49.5 billion to $32.3 billion that year. A D3 code that continues covers a smaller slice of IT work, so its score can move with no change in the market. For now each line stays within one code era, FY2008 to FY2011 (4 years), FY2012 to FY2020 (9 years) or FY2021 to FY2025 (5 years). The proposed rule calls a line rising when the average HHI of its last 3 years is at least 500 points above the average of its first 3. Averaging 3 years softens the effect of one large award, such as a multi-year contract recorded in a single year. A line needs 6 scored years so the two windows do not overlap, which puts every line in FY2012 to FY2020. It also needs $10 million in total. Small segments swing the most from year to year (Figure 9), so this floor drops the noisiest lines and still keeps 98.9% of the dollars. That leaves 5,772 trend lines, each one agency and product code followed over FY2012 to FY2020. Large segments swing too, though, so the floor cannot remove the noise on its own. Section 3.4 measures how much remains by testing how often the rule fires by chance. The 500-point cutoff is a judgment call. The 2023 guidelines treat a 100-point rise in a highly concentrated market as significant [15], but they mean the rise caused by one merger in a market that otherwise holds still. Our segments do not hold still. A segment's score can move hundreds of points in an ordinary year as large contracts start and end. A 100-point cutoff would call most of that ordinary movement a rise, so we use 500 and repeat the test at 250 and 1,000 as a sensitivity check. To measure what noise alone produces, we keep each line's yearly scores but put the years in random order. We do this 20 times with a fixed seed and apply the same rule to the shuffled lines.

3. Results

3.1. The data can support the score, with three weak spots

The copy is verified, and its yearly totals match GAO's within 0.2% for FY2022 to FY2025. In practice, the copy holds the same contract dollars the government's auditors report, so no recent year has lost or doubled a meaningful share of its records. The summary tables built for this analysis pass all 116 checks, listed in Appendix A, and in every year their rows and dollars add back to the verified copy. The proposed data handling changes the results noticeably only in FY2008 to FY2010, where the known bad records sit (Appendix B). Three weak spots remain.

One firm can look like two. We already roll each company up to its parent company ID. The parent field can be blank or lag a merger, though, so in some years a firm still appears under two or more IDs. One firm counted as two lowers HHI with no change in the market. In 81 cases, one firm appears under two or more IDs that each carry its own name in the same year. Lockheed Martin, General Dynamics and McKesson are among them, and most cases fall in FY2024. Optum and RTX also split, under IDs with different names. These splits are not later acquisitions read back in time. Each name-matched ID carries the firm's name on that same year's records. The integrity review also found that a row's parent ID is the parent at the time of the action, switching two to six months after a deal closes [9]. The reverse also happens. A joint venture puts two or more firms under one ID. That can make a market look more concentrated, or hide part of a member's share when the member also wins under its own ID. The next phase measures both (Section 4). The effect of splits on one market can be large. VA managed health care scores 6,991 in FY2024 with Optum's two IDs apart and 9,641 with them together. The effect on the total is small. Merging every split raises the FY2024 share of dollars above 2,500 by 1.6 points (Appendix E).

Returned money changes the measure little. Deobligations can make a vendor's total negative, and a segment where no vendor nets above zero gets no score (Figure 3). Those segments hold almost no money, so leaving them out barely moves the measure. Ignoring returned money altogether gives nearly the same share in every year, within 1.3 points (Figure 4). Most negative dollars take back money obligated earlier on the same award (Appendix F).

Two code changes split the years. The FY2012 change added codes mostly for construction and repair, and the FY2021 change added codes almost all for IT. Old codes kept dollars for years after each change, so lines stay within one code era until we test which codes kept their meaning. That limits question 2 to FY2012 to FY2020 for now. Codes the changes left alone could carry lines across all 18 years. Telling them apart from codes that kept their number but changed meaning, like D3, is a next step.

3.2. Half of civilian contract dollars sit in highly concentrated segments

FY2012 to FY2020 is the longest stretch of consistent product codes. Within it, the civilian share of dollars in highly concentrated segments rose from an average of 41% in its first three years to 44% in its last three (Figure 1). Defense rose more over the same years, from 40% to 46%. The civilian share has averaged 49% since FY2021. Part of the step at FY2021 may come from that year's code change, which split the IT codes into finer ones and can raise HHI with no change in the market. We have not yet measured how much, and a next step scores the new IT codes grouped back together.

The money grew too. Civilian contract spending roughly doubled from FY2008 to FY2025, a rise of about 43% after inflation, and services account for most of the growth

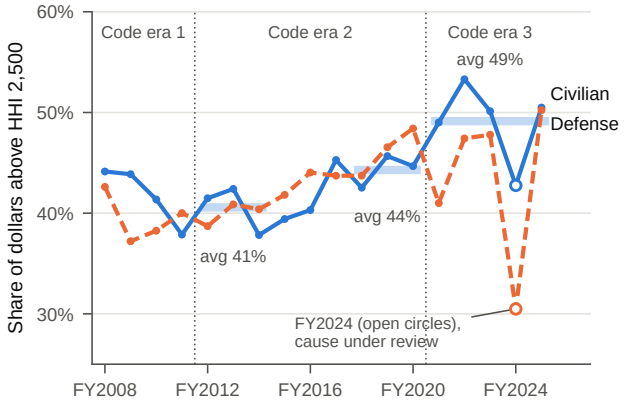


Figure 1. Civilian concentration rose within the longest code era and has stayed higher since FY2021. Share of contract dollars in segments with HHI above 2,500, by fiscal year, under the proposed data handling, for civilian agencies (solid) and Defense (dashed). Dotted lines mark the two product code changes. The light blue bars mark the civilian average over FY2012 to FY2014, FY2018 to FY2020 and FY2021 to FY2025. FY2024 (open circles) dips. Section 3.2 finds where the drops sit, and their cause is still under review.

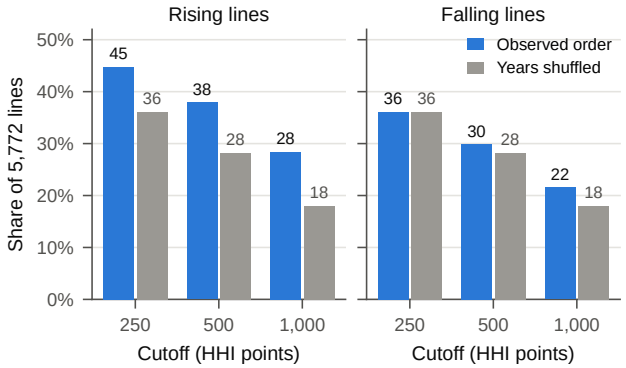


Figure 2. At every cutoff, many lines still rise after shuffling, so most apparent rises could be chance. Share of the 5,772 trend lines (FY2012 to FY2020) whose last 3 years average at least the cutoff above their first 3. Gray bars use the same lines with their years in random order.

Table 2. Three VA health segments are each held almost entirely by one company. The five largest civilian segments with HHI above 2,500 in FY2025, under the proposed data handling. VA is Veterans Affairs and DHS is Homeland Security. Share is the top vendor's share of the segment's dollars, and Firms counts the vendors with a positive net. TriWest's one rival holds under 0.01% of its segment.

Segment	$B	Top vendor	Share	HHI	Firms
VA managed health care	22.8	UnitedHealth	98.4%	9,680.2	75
VA drugs and biologicals	11.7	McKesson	98.0%	9,606.1	193
VA health insurance	9.5	TriWest	>99.9%	9,999.8	2
VA medical screening	8.1	Leidos	34.4%	2,862.8	36
DHS non-building facilities	5.1	BCCG JV	60.9%	4,276.6	21

(Figure 5). In FY2025, about half of it, $152 billion, sat in segments above 2,500.

The FY2024 dips need care. Most of the civilian dip comes from one Department of Energy segment, the operation of miscellaneous buildings (code M1JZ), whose score fell below 2,500 (Figure 6). The Defense dip spreads over several large segments, led by combat ships, that drop in FY2024 and recover in FY2025. Vendor ID splits explain little of either dip (Appendix F). Another candidate is contract timing, the subject of Section 3.4. No trend claim should rest on FY2024 until its cause is found.

3.3. Concentration sits in a few places

Veterans Affairs (VA) is the largest civilian buyer. Three quarters of its FY2025 dollars sat above 2,500, up from a third in FY2008 (Figure 7). Its three largest concentrated segments are each held almost entirely by one company (Table 2), and together they hold about 15% of civilian contract dollars. The vendor count shows two kinds of concentration. Health insurance has two vendors, so it is concentrated by arithmetic. Managed health care and drugs have 75 and 193 vendors, so one firm dominates many small ones. Across all agencies and years, segments with one to three vendors hold under 5% of the dollars, so most concentrated money sits where one firm leads many (Appendix F). Among the ten largest civilian buyers, NASA is the only other one with more than three quarters of its dollars above 2,500. Single product codes can also overstate change. VA's code Q201, now labeled managed health care, grew almost sixty times from FY2019 to FY2025, but part of that growth is money moved over from another VA code in FY2021 (Figure 8).

3.4. Yearly noise is large next to a 500-point rise

Shuffling keeps each line's scores but scrambles their order, as if the years were dealt like cards. Any real trend disappears, while the size of the year-to-year swings stays. At the 500-point cutoff, 38% of real lines rise, and 28% of shuffled lines still rise (Figure 2). So chance alone produces about three quarters as many rising lines as the data shows. Falling lines barely beat chance. The same pattern holds at 250 and 1,000 points. That is because scores move a lot from one year to the next. Even among the largest tenth of segments, 42% of yearly changes are 500 points or more (Figure 9). Lines that start high also tend to fall, as regression to the mean would predict (Figure 10). Part of that movement may be chance, and part may be contract timing. A score can jump in the year a large contract is signed and fall back after it, a cycle the shuffle breaks along with any trend. For any single line the rule cannot yet tell a real rise from chance or from a contract cycle. Spreading each contract's dollars over its period of performance, the dates the work runs, would separate the two. The rule has to change before question 3 compares features.

3.5. Concentrated money carried a competed code as often as the rest

Concentrated civilian dollars were competed about as often as the rest. Across all 18 years, 85.3% of concentrated dollars and 86.3% of other dollars that carry a competition code were competed. In FY2008 to FY2015, concentrated money was competed less often, by up to 6.5 points. From FY2016 on, the two rates have crossed back and forth (Figure 11). The three large VA segments were competed on 99.6% to 100% of their dollars. NASA is the exception among large buyers, with 36.8% of its concentrated FY2025 dollars awarded without competition. The pattern fits agencies that run one competition and award a whole market to one firm. For oversight, the open question is how many firms bid, and the competition code does not record that. Much of this money flows through umbrella contracts. An agency first competes an umbrella contract that names one or more approved suppliers, then places orders under it for years. An order's record may say it was competed because the umbrella was, even when the order itself went to the only supplier on it. Orders under umbrella contracts hold 72.7% of concentrated civilian dollars in FY2025, so this matters. We still have to confirm in the official data definitions which award the code describes.

4. Discussion

The score holds up. USAspending records can support a concentration score once a short list of known problems is handled. Most of that money carries a competed code, so sole-source awards do not explain most of it. For orders, though, the code may reflect the umbrella contract's competition (Section 3.5). One likely reading is that agencies award large, market-wide contracts through competition, which the next phase will test. Single years mislead, through chance, contract timing and data problems, so trends must rest on several years.

Oversight offices can find one-firm markets. Offices that oversee federal buying, such as GAO and agency competition advocates, can use the measure to find where an agency depends on one firm even after a competition. VA health care is the clearest case. There the useful question is whether the next competition draws real rivals.

One question answered, one started. Question 1 is largely answered. Question 2 has started. Its first result shapes the method. A 500-point rise over a 9-year line often looks like chance, and we do not yet know how much of the swing is contract timing. The trend lines also end in FY2020, so question 2 cannot yet speak to the higher shares since FY2021. Question 3 waits until the rising rule is fixed.

The plan narrowed to civilian agencies. We replaced the class pitch's plan to use the USAspending API with the full archive files, and we dropped its inflation adjustment, which HHI does not need. The first results moved the focus from every agency to the civilian agencies, since DoD already reports on its own supplier base [2].

We would check vendor IDs first. The pitch named companies hiding inside other companies as the first risk, yet the hand check of the 100 largest vendors was scheduled after the first results. Checking vendor IDs year by year ahead of the trend work would have caught the FY2024 splits earlier.

Next steps.

1. Merge split vendor IDs, starting with the 81 firms found by name, and check the 100 largest vendors by hand. Test joint ventures against their member firms.
2. Spread each contract's dollars over its period of performance, after cleaning the few impossible dates, to test whether yearly swings come from chance or from contract timing.
3. Track concentrated dollars separately for segments with few vendors and segments where one firm leads many.
4. For question 2, let lines cross a code change where the code kept its meaning, so they reach the years after FY2020. Score the new IT codes grouped back together to measure the FY2021 step. Sort the lines into a few shapes with k-means clustering. Set a rising rule that beats the shuffle by a wider margin, varying the cutoff, the window length and the size floor together.
5. For question 3, compare rising lines with the rest using an L1-penalized logistic regression and a shallow decision tree. Fill in offers received, the number of bids an award drew, beyond FY2012 to FY2017, the years it is recorded consistently.
6. Test, as stretch goals, whether competition stays real when a concentrated market's contract is awarded again. Also check VA health care one level down with subcontract data, and score each product code across all agencies.

5. Statement of contributions

Suchitra Hole, Shivram Nekkanti and Andy Hannah contributed equally to this project.

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Appendix

A. Code and data

All code queries the verified copy with DuckDB from Python. The pipeline is in `src/` on the main branch of the GitHub repository. The exploratory tables and their code are in `reports/eda/` and `src/eda/` on its `shivram-eda` branch. The figure script and the checks behind Section 3.1 and Appendices D and E will be added to the repository before the final report. The data files are too large for the repository and come from the Award Data Archive [8].

The 116 checks. Six checks run for each of the 18 years, 108 in all. Each year's rows and net dollars equal the verified copy. Its segments, vendor totals and negative vendor totals equal the counts in the integrity report. Every split of the year's data adds back to its total. Four checks for each of FY2024 and FY2025, 8 in all, confirm that those years as published reproduce the proposal's preliminary results. All 116 pass (`reports/eda/base_checks.csv`).

B. Proposed data handling against the data as published

The proposed handling matters only in FY2008 to FY2010, the three years with known bad records. It lowers the share of dollars above 2,500 by 0.97 to 3.65 points in those years and moves it by 0.17 points at most in every later year. There are 365,402 segments as published and 357,284 under the proposed handling. Share of dollars in segments with HHI above 2,500, all agencies.

Fiscal year	Data as published	Proposed handling	Difference	What moves it
FY2008	46.69%	43.04%	−3.65	The $13.9 billion entry reversed in FY2009, and the foreign placeholder
FY2009	40.20%	39.23%	−0.97	The reversal of that entry
FY2010	41.51%	39.25%	−2.26	The two $10 billion US Mint rows
FY2011 to FY2023	Within 0.17 points of each other		0.17 at most	Small effects only
FY2024	35.52%	35.52%	0	None
FY2025	50.33%	50.33%	0	None

C. The data decisions

The integrity review proposed nine data decisions, and a later review of the repository added two. None is final yet. Four of them change the data in this report, the ones Section 2 lists under proposed data handling. The decision log in the repository records each decision and how this report handles it (`config/decisions.csv`).

D. Large amounts recorded and then reversed

Pairs that cross a fiscal year are the ones that move the measure, since the two rows of a same-year pair largely cancel out. The integrity review handled five pairs [9]. Among them only the Emerson pair crosses a fiscal year, and its $13.9 billion entry turns one Defense construction segment's HHI from 167 into 4,282 in FY2008. Four of those five came from a table of the ten largest rows across all 18 years. The fifth, a Navy pair that nets to $0.00, ranks 21 and 22. The cut at ten left out the KEPA-TCI and Lawrence Livermore pairs, whose rows rank 11 and 12, and 17 and 18. A full scan of every row finds 116 pairs of $100 million or more on the same award, with opposite signs and amounts within 1% of each other. Of these, 75 look like entry errors and 41 like normal accounting, such as a closeout, and 66 are marked for review by hand. The proposed handling nets only the five pairs the integrity review found. KEPA-TCI and Lawrence Livermore are not netted, but each pair's two rows fall in the same year, segment and vendor, so they cancel and leave the score unchanged. CBY Design Builders, a Defense construction joint venture, crosses a year and is not yet netted, so it is the next pair to fix. It touches one Defense segment in FY2011 and FY2012, and no civilian figure.

Vendor	Agency	Amount	Entry and opposite entry	Fiscal years	In this report
Emerson Construction	Army	$13.9 billion	2008-09-29 and 2009-01-07	FY2008 to FY2009	Netted
CBY Design Builders	Army	$0.7 billion	2011-04-13 and 2012-09-28	FY2011 to FY2012	Not yet netted
KEPA-TCI JV	Army	$10.7 billion	2015-07-29 and 2015-07-31	FY2015 only	Cancels within the segment
Lawrence Livermore National Security	Energy	$7.9 billion	2012-07-25 and 2012-07-31	FY2012 only	Cancels within the segment

E. Vendor ID splits

Improvements for the next round. Across FY2008 to FY2025, 81 large firms appear under two or more vendor IDs that carry the same name in the same year. The splits cluster in the last few years, and FY2024 alone holds 51 of these firms. In FY2024 the IDs other than each firm's largest carry $48.7 billion. Counting each split firm as one vendor raises the FY2024 share of dollars in segments with HHI above 2,500 by 1.6 points for all agencies. In VA managed health care it raises the FY2024 HHI from 6,991 to 9,641. The table shows how the money splits for five firms in FY2024. Each ID beyond a firm's largest is scored as a separate vendor, so the share on those IDs is concentration the measure misses.

Firm, FY2024	IDs	Largest ID	Other IDs	Share on other IDs
Lockheed Martin	12	$32.2B	$17.6B	35.3%
General Dynamics	3	$10.1B	$5.9B	36.8%
McKesson	2	$6.5B	$1.7B	21.1%
Optum (UnitedHealth)	2	$18.8B	$3.3B	15.0%
RTX (Raytheon)	2	$23.7B	$1.3B	5.2%

F. Supporting figures for Sections 3 and 4

The main text gives one headline number per point. Figures 3 to 11 show the series behind those points, each with its message first. Table 3 holds single figures. Paths are under `reports/` in the repository. "Review" marks a figure recomputed in the independent review of this report, in `docs/reviews/2026-10-24_report_v1/recomputed.csv`.

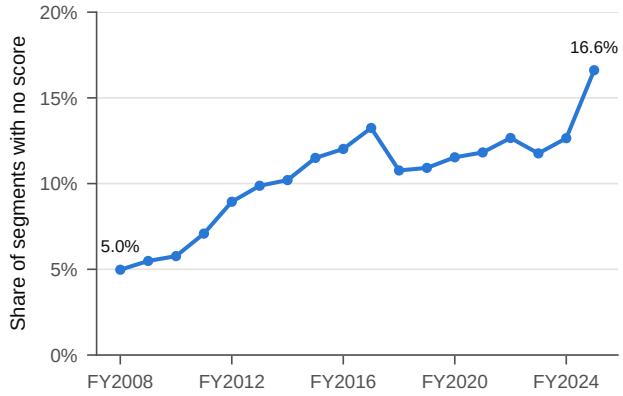


Figure 3. More segments go unscored over time, but they hold almost no money. Share of segments with no score, where no vendor nets above zero, by fiscal year. These segments hold 0.07% of a year's dollars at most. Source: `eda/concentration_by_year.csv`.

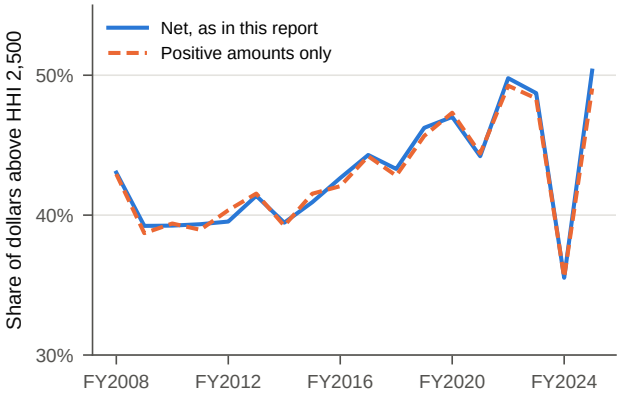


Figure 4. Ignoring returned money barely changes the measure in any year. Share of contract dollars in segments with HHI above 2,500, all agencies, from net amounts, as in this report, and from positive amounts only. The largest gap is 1.3 points, in FY2025. Source: `decision_evidence/p3_measures_by_year.csv`.

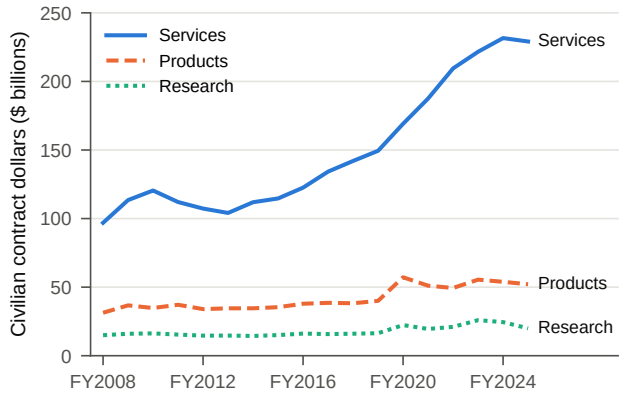


Figure 5. Services drive almost all of the growth in civilian contract spending. Civilian contract dollars by kind of purchase, in current dollars. The total grew from $143.3 billion in FY2008 to $301.2 billion in FY2025, 43% more after inflation [16]. Services grew from $96.9 billion to $229.1 billion, 84% of the growth. VA managed health care alone grew by $22.7 billion over the same years, about a sixth of the services growth. Source: `eda/civilian_by_kind.csv`.

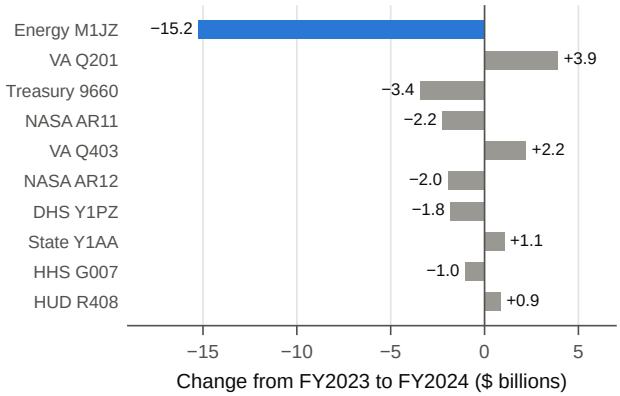


Figure 6. One Energy segment accounts for most of the FY2024 civilian drop. Change in civilian dollars in segments above HHI 2,500 from FY2023 to FY2024, for the ten segments that changed most. Labels give the agency and the product code. Energy's M1JZ, the operation of miscellaneous buildings, accounts for $15.2 billion of the $19.4 billion drop. Source: `eda/civilian_drivers_fy2024.csv`.

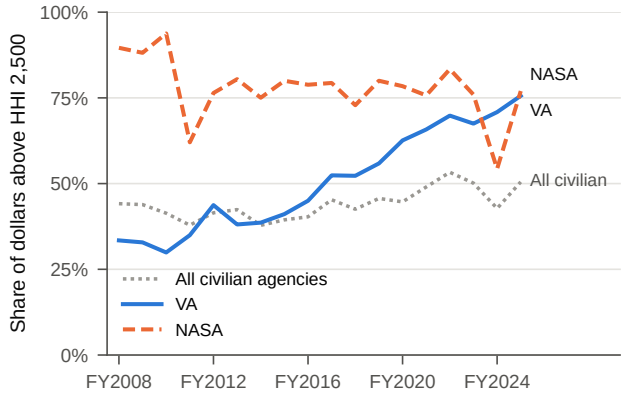


Figure 7. VA's concentration more than doubled, while NASA stayed high in most years. Share of each agency's contract dollars in segments with HHI above 2,500. VA held $78.3 billion in FY2025, with 75.6% above 2,500, against 33.4% in FY2008. NASA held $21.3 billion, with 77.0% above 2,500. Source: `eda/civilian_by_agency.csv`.

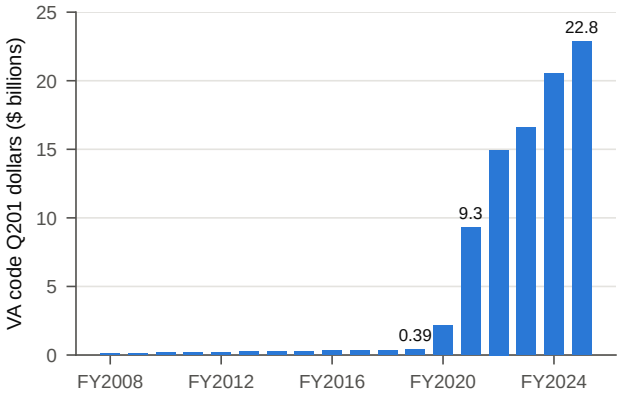


Figure 8. VA's managed health care code was small until FY2020, then grew to almost sixty times its FY2019 size. VA contract dollars under code Q201, now labeled managed health care. In FY2021, VA's "medical, other" code (Q999) fell by $5.9 billion while Q201 rose by $7.1 billion, so part of the growth is money moving between codes. Source: `eda/example_segment.csv`, `eda/drivers_above_2500.csv`.

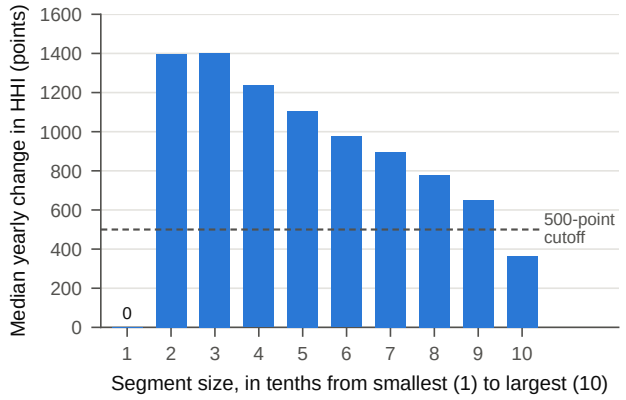


Figure 9. Smaller segments swing more, but even the largest swing often. Median yearly change in a segment's HHI, by segment size in tenths of all segment-year pairs. In the largest tenth, above about $27 million, the median change is 364 points, and 42% of changes are 500 points or more. Source: `eda/noise_by_size.csv`.

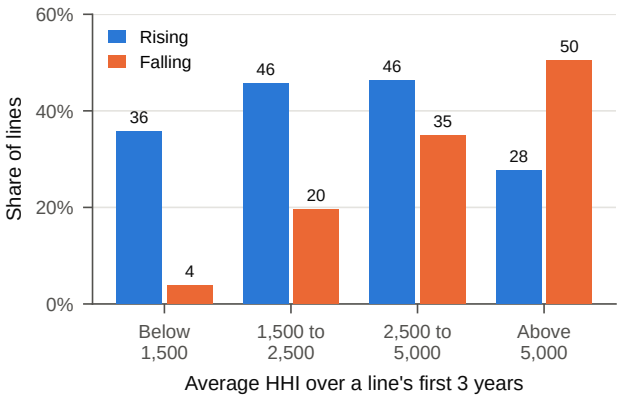


Figure 10. Lines that start high fall more often than they rise, as regression to the mean predicts. Share of trend lines that rise or fall by 500 points or more, by the average HHI of their first 3 years. Lines that start above 5,000 fall more often than they rise, 50.4% against 27.7%. Source: `eda/rising_by_start.csv`.

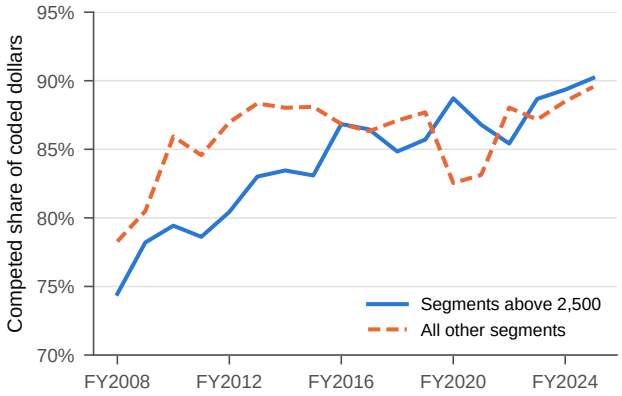


Figure 11. Concentrated money was competed less often than other money before FY2016, by up to 6.5 points, and the two rates have crossed back and forth since. Both rates rose from about three quarters to about nine tenths. Shares of the dollars that carry a competition code. Across all 18 years, 85.3% of concentrated and 86.3% of other dollars were competed, and 14.3% and 13.4% were sole source. Source: `eda/civilian_competition_by_year.csv`.

Table 3. Single figures behind Sections 3 and 4, with their sources.

Section	Point	Figures	Source
3.1	Proposed handling moves few segments between four bands, 1,500 or less, above 1,500 to 1,800, above 1,800 to 2,500 and above 2,500	0.9% of all segments change band, 0.65% to 1.13% by year. From FY2011 on, the share above 2,500 moves 0.17 points at most.	`eda/band_changes.csv`, Appendix B
3.1	Optum's two IDs in VA managed health care, FY2024	82% and 16% of the segment, the second ID $3.31 billion	`decision_evidence/p1_va_check.csv`
3.1	Share above 2,500 in FY2024 after merging split IDs	All agencies 35.5% to 37.1%. Civilian 42.8% to 43.2%. Defense 30.5% to 32.9%.	`decision_evidence/p1_merge_effect_by_year.csv`, Review for Defense
3.1	Negative totals	7.5% of vendor totals within a segment are negative. Leaving unscored segments out moves the share above 2,500 by 0.08 points at most.	Integrity review [9], Review
3.1	Negative dollars on awards with earlier positive obligations	90.7% over all years	`decision_evidence/p4_negative_dollars_by_source.csv`
3.1	Dollars on new product codes	6.5% of FY2012 dollars and 4.4% of FY2021 dollars	`eda/psc_break_fy2012.csv`, `eda/psc_break_fy2021.csv`
3.2	Civilian peak	53.3% in FY2022	`eda/concentration_dod_civilian.csv`
3.2	FY2024 Defense drop	17.3 points from FY2023, against 2.4 points regained by merging split IDs	`eda/concentration_dod_civilian.csv`, Review
3.3	Segments with one to three vendors	4.9% of scored dollars across all agencies and years, against 43.4% of dollars above 2,500	`eda/hhi_by_vendor_count.csv`, `eda/concentration_dod_civilian.csv`
3.3	Three VA segments in Table 2	$44.0 billion, 14.6% of civilian dollars, top vendor 98% to 100%	`eda/civilian_top_segments.csv`
3.4	Rising lines at the 500-point cutoff	37.9% observed and 28.2% shuffled, a gap of about 560 of 5,772 lines	`eda/rising_counts.csv`
3.2	Jump from FY2024 to FY2025, all agencies	14.8 points, and 13.6 points after merging split IDs	`decision_evidence/p1_merge_effect_by_year.csv`